Venkata Sai Surya Rao Beeraka

Software Engineer

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SUMMARY

Software Engineer with 3 years of experience developing and scaling mission-critical microservices. Proven track record integrating AI-powered LLM/RAG workflows that cut research time by 30% and implementing containerized CI/CD pipelines for on-demand cloud deployment. Expert in building resilient event-streaming layers, optimizing complex database pipelines, and engineering backend systems that double operational throughput. Collaborative Agile engineer who pairs modern frontend practices with robust infrastructure to deliver high-performance enterprise platforms.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL (T-SQL, PL/SQL), HTML5, CSS3/SASS
Frameworks & Libraries: Spring Boot, Angular, Node.js, React, Enterprise Integration Patterns (EIP)
Cloud & Infrastructure: Docker, Kubernetes, Jenkins, CI/CD Pipelines, AWS/Azure, Microservices Architecture
Data & Streaming: Apache Kafka / RabbitMQ (Event Streaming), SSIS (ETL), MS SQL Server, Oracle DB
AI & Applied Technologies: LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering
Tools & Methodologies: Git, Jira, Agile/Scrum, RESTful API Design, Unit/Integration Testing (JUnit, Jasmine)

EXPERIENCE

UIC College of Medicine | **Jun 2025 – Dec 2025| Research Software Engineer**

- Supported psychiatric AI research by applying machine-learning methods to early detection and risk stratification across 5,000+ de-identified patient records.
- Developed NLP workflows to analyze 12,000+ clinical notes and patient narratives, reducing manual research-review time by 25%.
- Evaluated bias and fairness across 8 model iterations and 4 demographic cohorts, identifying 3 key performance disparities to improve model reliability.
- Built reproducible Python-based data-preprocessing pipelines that improved research-data quality by 20% and reduced model-training preparation time by 30%.
- Partnered with biostatisticians and clinical researchers to validate 6 predictive models, achieving up to 87% classification accuracy for mental-health risk identification.

Carelon Global Solutions, India | Oct 2022 - Jul 2023 | Software Engineer

- Designed and built a comprehensive healthcare insurance platform enabling claims processing, patient data management, and XML feed ingestion, streamlining core operational workflows.
- Collaborated with product stakeholders to design and implement key features, including patient data dashboards and broker management modules using Angular, directly improving user navigation.
- Optimized ETL pipelines using T-SQL and SSIS, significantly improving claims processing performance and reducing overall runtime.
- Maintained high code quality standards by writing comprehensive unit and integration tests to prevent production regressions and ensure platform reliability.

Carelon Global Solutions, India | Aug 2021 - Sep 2022 | Associate Software Engineer

- Spearheaded the development of the ACE-T (Agent Commission Engine – Tiering) platform using Java and Spring MVC to manage agent commission workflows, tiering calculations, and payout operations across internal web portals.
- Designed and maintained RESTful APIs with an Oracle PL/SQL backend, optimizing commission processing workflows, improving data consistency, and successfully recovering lost revenue from past agent overpayments.
- Established Jenkins CI/CD pipelines for automated Spring Boot service deployments, improving release speed and operational reliability.
- Participated in 100+ Agile sprints using Jira, collaborating closely with QA teams and project managers to deliver platform enhancements on schedule.

PROJECTS

CollabDocs

- Designed and implemented a resilient real-time collaborative editor using CRDTs for multi-user/offline sync and streamlined real-time synchronization by deploying a Dockerized WebSocket backend on AWS ECS, reducing lag by 300ms.
- Engineered extensible collaboration infrastructure using Firestore versioning and Yjs state merging, achieving 42% memory savings and seamless cross-device sync, with the project open-sourced to foster accessibility and community adoption.

Scalable Payment Gateway

- Designed and deployed a fault-tolerant payment gateway using Java, Spring Boot microservices and built a Kafka stream-based fraud detection engine, improving fraud flagging accuracy to 98% and reducing detection latency by 40%.
- Integrated Prometheus-based observability into the pipeline, enabling real-time tracking of transaction throughput, fraud alerts, and service health, with automated alerting and dashboards improving incident response time by 60%.

BeyondSonar

- Built a Python Flask microservice that automatically generated security reports from SonarQube Hotspot data using Retrieval Augmented Generation and Llama 3.1 8B, improving vulnerability analysis and accelerating security triage for developers and managers.
- Designed an LLM powered reporting pipeline using Qdrant vector search, Sentence Transformers, and OWASP security guidance to generate accurate vulnerability analysis, remediation recommendations, and business focused PDF reports with over 90% retrieval accuracy.

CERTIFICATION

AWS Certified Cloud Practitioner

EDUCATION

Master of Science in Computer Science | University of Illinois at Chicago, USA — **May 2025**
Bachelor of Technology in Computer Science and Engineering | Jain University, Bangalore, India — **May 2021**